

Energy consumption for business travel at KfW Group¹ in megawatt-hours

	2014	2015	2016	2017	2018
Car (petrol and natural gas)	200	263	659	416	428
Car (diesel)	6,725	7,391	5,071	5,953	5,506
Train	968	944	1,095	1,248	1,167
Flights (Germany)	5,027	4,409	4,839	4,626	4,785
Flights (Europe)	2,937	2,374	2,172	2,587	2,481
Flights (international)	25,850	19,809	19,558	22,235	23,799
Total	41,707	35,190	33,394	37,065	38,166
Per capita	6.6	5.5	5.0	5.0 ²	5.0

¹ Calculated according to figures from the Association for Environmental Management and Sustainability in Banks, Savings Banks and Insurance Companies (VfU); factors including upstream chain

² The system limit for surveying employees was changed in 2017, making it impossible to perform direct comparisons with per-capita consumption in previous years.

Measures for reducing energy consumption

In the context of a project for the energy-efficient restructuring of the central heating and cooling systems at the Frankfurt site, KfW completely restructured the system between the end of 2016 and spring 2019. This project was a response to the changing consumption structures within the building complex and to the challenges of the energy transition.

The measure was and is aimed at establishing a centralised, flexible and energy-efficient energy supply to the location that is sustainable in the long term. The solution uses large-scale heat pumps and combined heat and power systems to supply heating. Furthermore, waste heat from the data centre in the Westarkade building in Frankfurt will be used to heat office space in the future. The new heat pumps will raise the low temperature in the data centre to a generally usable level,

enabling it to supply a portion of the entire heating network. In respect of cooling, the company launched a highly efficient turbocore chiller in 2018. The combined heat and power system has been running on a trial basis since the end of the year under review. Altogether, these measures will enable carbon emissions to drop by up to 1,500 tonnes per year.

In 2017/2018, part of the existing uninterruptible power supply (UPS) system was exchanged for modern, smaller models in an additional project. Efficiency gains of 220 megawatt-hours of electricity per year are achieved as a result of the reduction in power loss.



In the future, the Frankfurt site will be able to save up to 1,500 tonnes of CO₂ each year

Emissions

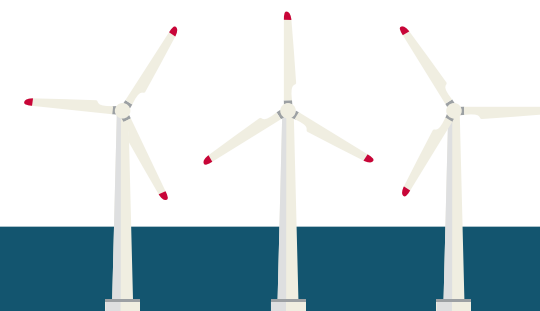
KfW Group's banking operations generate carbon emissions as a result of the consumption of energy and heat, the use of fuel during business trips, and due to events. The goal is to ensure climate-neutral banking operations wherever possible.

Climate-neutral banking operations

KfW Group's banking operations are largely climate-neutral in the areas of energy (buildings) and business trips. KfW offsets the emissions that remain despite the conversion to renewable energy sources and the use of more environmentally friendly transportation. Furthermore, it also assigns a CO₂ factor to the total green energy it consumes, as the use of green power still generates emissions. Electricity consumption at the external data centre is also taken into account in the compensation process. So far, not all avoidable emissions have been completely offset. KfW Group continually strives to record and offset as many emissions as possible.

To offset its carbon emissions, KfW Group purchases high-value Clean Development Mechanism (CDM) certificates on the market as well and discontinues them for good. In 2018, the majority of these certificates originated from a wind power project in India.

When evaluating its carbon equivalents (CO₂e), KfW uses international standards set by the Greenhouse Gas Protocol, and emissions are divided into Scopes 1 to 3. Scope 1 fell marginally by 0.8%. The rises under Scope 3 were caused by increased travel and events. The unusual 38.7% increase in Scope 2 was the result of increased use of heating oil and diesel-powered emergency generators due to building work. Nevertheless, the amount of CO₂e emitted per capita remained constant overall.



In 2018, CO₂ emissions were primarily offset by certificates for a wind power project in India

CO₂e emissions of KfW Group¹ in tonnes

	2014	2015	2016	2017	2018
Emissions from direct energy consumption (Scope 1) ²	4,636	5,095	5,260	5,616	5,571
Emissions from indirect energy consumption (Scope 2) ³	2,073	1,097	1,400	1,368	1,898
Total business travel (Scope 3) ⁴	9,446	7,394	7,411	8,208	8,665
Events (Scope 3) ⁵	102	102	74	87	101
Total	16,257	13,628	14,145	15,278	16,236
Per capita	2.6	2.2	2.1	2.1 ⁶	2.1 ⁶

¹ Unavoidable CO₂e emissions at the KfW Group have been offset since 2006.

² Natural gas, wood pellets and own fleet

³ Green electricity, district heating, oil and emergency power generators (diesel)

⁴ Only flights; short and long-distance rail travel is CO₂e-neutral per se.

⁵ Emissions for participants' travel to and from KfW buildings for events

⁶ The system limit for surveying employees was changed in 2017, making it impossible to perform direct comparisons with per-capita consumption in previous years.

Location- and market-based CO₂ emissions of KfW Group according to Scope 2 (2017) in tonnes

	Location-based, 2017	Market-based, 2017	Location-based, 2018	Market-based, 2018
Green electricity	5,869	4,636	5,819	4,596
Emergency power generators (diesel)	2,555	2,073	3,577	2,902
Heating oil	8,113	9,446	80	80
District heating	55	102	88	163
Total	16,592	16,257	9,564	7,732

According to the Greenhouse Gas (GHG) Protocol, we identified the Scope 2 emissions (from green electricity, district heating, oil and diesel emergency power generators) according to two different factors: location-based (average factor for Germany) and market-based (factor for the energy supplier) factors.

This data has been recorded since 2017.

Resource consumption

Paper is one of the essential materials of a financial institution. KfW Group continuously strives to reduce consumption. This is achieved, for example, by increasing employee awareness of the environmental impacts of paper consumption and

by switching from paper-based to digital processes in specific areas. Despite an increasing level of digitisation at KfW, the total amount of paper consumed increased by around 9% in the year under review. Per capita consumption rose by around 2 kilograms in total.

By far, the largest percentage of paper used at KfW Group – nearly 84% – is recycled paper with the “Blue Angel” environmental label. Publications intended for public use are printed exclusively on FSC® paper, which means that we do not use any ECF, TCF or fresh-fibre paper bleached with elemental chlorine. Furthermore, the group only works with climate-neutral printing companies – a further contribution towards climate neutrality at KfW.

84%

of KfW Group's office
paper bears the
“Blue Angel” label

KfW Group paper consumption in Germany in kilograms

	2014	2015	2016	2017	2018
Recycled paper (Blue Angel) ¹	308,385	251,150	225,863	196,084	213,000
Writing paper (recycled paper – Blue Angel) ¹	5,260	3,351	3,171	2,635	1,865
Virgin fibre paper for promotional materials (FSC mixed sources label)	72,931	153,512	73,291	32,528	39,704
Total	386,576	408,413	302,325	231,247	254,569
Per capita	61.5	64.3	45.5	31.2 ²	33.3
of which, recycled paper (in %)	81	62	76	86	84

¹ Refers to 80 g/m² A4 printer and copy paper.

² The system limit for surveying employees was changed in 2017, making it impossible to perform direct comparisons with per-capita consumption in previous years.

Water management

According to the sustainability guidelines for in-house environmental protection, KfW Group strives to continually reduce water consumption within the group. Existing public infrastructure is used for waste water. Separation systems make sure that the relevant statutory limits are met.

Water consumption

Up to around 84% of water is drawn from the municipal pipeline network. The remaining 16% comes from ground water (in Berlin, ground water or stratum water occurring at underground garages is used) and collected rainwater (process water). The latter is used for outdoor irrigation at the Frankfurt, Berlin and Bonn locations and, in Bonn, it is also used for flushing toilets. The ground water in Berlin is used for toilet flushing and operating the cooling tower. After four years on a downward trajectory, per capita water consumption had more or less stagnated in 2018.

14.8

cubic metres of water
per capita were
consumed in 2018

KfW Group water use in Germany in cubic metres

	2014	2015	2016	2017	2018
Drinking water	94,209	98,808	95,830	94,819	95,502
Other process water (rainwater and ground water)	14,231	17,355	18,313	12,912	17,750
Total	108,440	116,163	114,143	107,731	113,252
Per capita	17.2	18.3	17.2	14.6 ¹	14.8

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